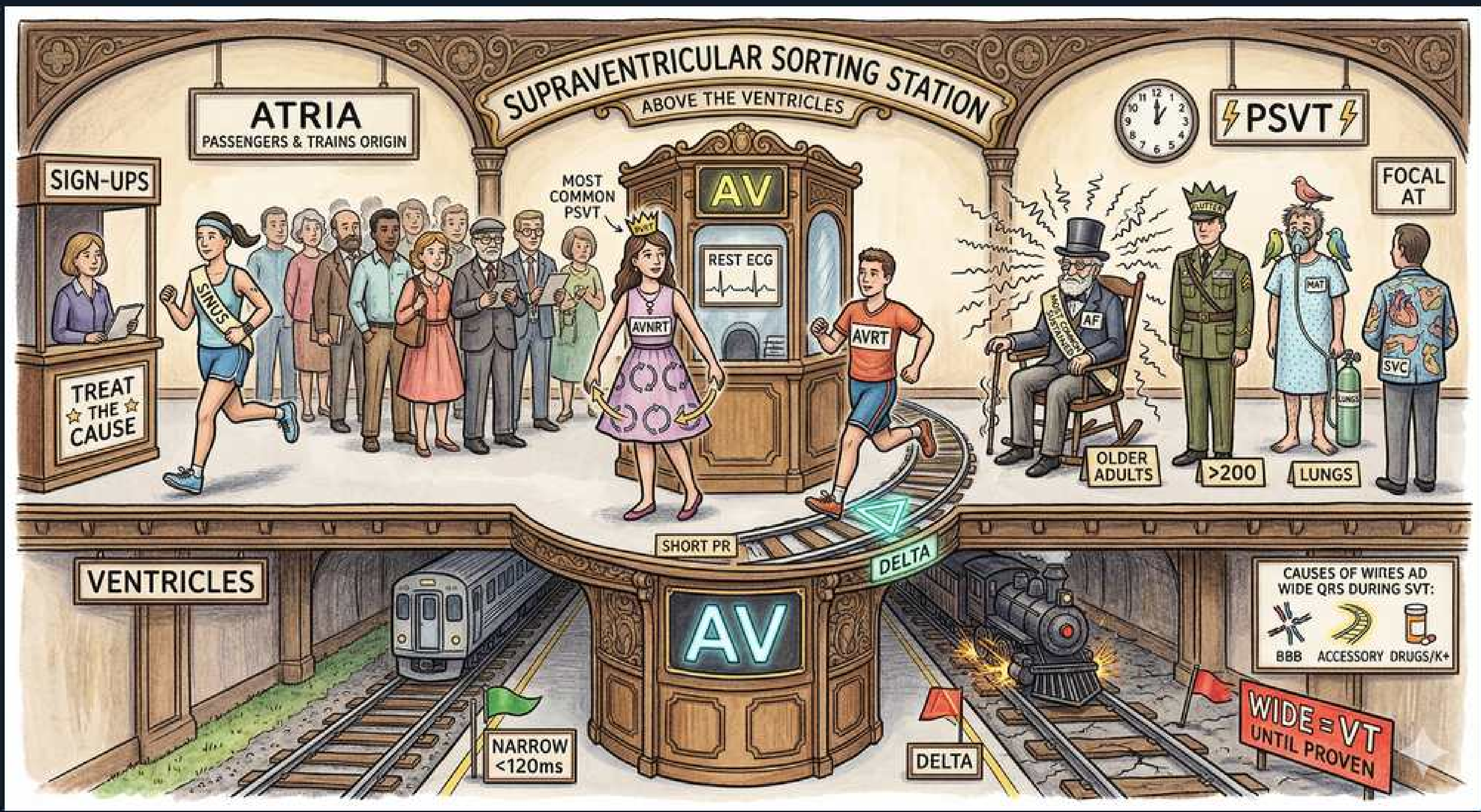
Arrhythmias — Approach to Supraventricular Arrhythmias



1. Atria above

WHAT YOU SEE

An upstairs platform crowded with passengers under an 'ATRIA' banner — the upper-level crowd = impulses born at or above the AV node (in the atria), the source of all supraventricular rhythms.

WHAT IT ENCODES

Supraventricular means the arrhythmia originates AT or ABOVE the AV node — in the atria or AV junction. By convention this produces a narrow QRS (<120 ms) because the ventricles are still activated through the normal His-Purkinje system. The major SVTs are sinus tachycardia, atrial fibrillation, atrial flutter, atrial tachycardia (focal/multifocal), AVNRT, and AVRT.

BOARD TRAP

'SVT' on a monitor is often used loosely to mean a regular narrow-complex paroxysmal tachycardia (PSVT = AVNRT/AVRT/AT), but technically AF and flutter are also supraventricular — be precise about which the question is testing.

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2. AV booth gatekeeper

WHAT YOU SEE

A ticket-control booth marked 'AV' on the platform between the upper and lower levels — the gatekeeper booth = the AV node, the obligatory checkpoint every atrial impulse must pass through to reach the ventricles.

WHAT IT ENCODES

The AV node is the gatekeeper that conducts (and physiologically delays) impulses from atria to ventricles. It is the obligatory limb of the re-entry circuit in AVNRT and AVRT, which is exactly why slowing/blocking the AV node (vagal maneuvers, adenosine, beta-blockers, non-dihydropyridine CCBs) terminates those arrhythmias.

BOARD TRAP

AV-nodal blockade terminates AV-node-DEPENDENT rhythms (AVNRT, AVRT) but only transiently slows AV-node-INDEPENDENT rhythms (atrial flutter, atrial tachycardia, sinus tach), which resume after the drug wears off — and is dangerous in pre-excited AF (WPW).

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3. Narrow QRS <120ms

WHAT YOU SEE

A 'NARROW <120ms' sign on the platform — a slim, narrow QRS means the ventricles fired normally through the His-Purkinje wiring, proving the rhythm came from above (supraventricular).

WHAT IT ENCODES

A narrow QRS (<120 ms / <3 small boxes) indicates the ventricles are depolarized via the normal His-Purkinje system, so the rhythm is supraventricular in origin. Next, characterize it by regularity and by the P-wave/RP relationship: irregularly irregular with no P waves = AF; sawtooth = flutter; regular with hidden or pseudo-r' P waves = AVNRT.

BOARD TRAP

Narrow complex does NOT automatically mean sinus rhythm — it can be AVNRT, AVRT, focal AT, flutter, or AF. Conversely, a wide QRS does not rule out SVT (aberrancy/BBB/pre-excitation can widen it).

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4. AVNRT most common

WHAT YOU SEE

A crowned princess labelled 'MOST COMMON PSVT' — the crown marks AVNRT as the queen (most common) of the paroxysmal SVTs, classically in young women.

WHAT IT ENCODES

AV nodal re-entrant tachycardia is the MOST COMMON paroxysmal SVT (~60% of PSVT), typically in young women. It uses dual pathways within/near the AV node (a slow and a fast pathway) that form a micro-reentry loop. On ECG the retrograde P wave is buried in or just after the QRS, producing a pseudo-r' in V1 or a pseudo-S in inferior leads (short RP tachycardia). Rate is typically 150-250 bpm.

BOARD TRAP

AVNRT, not AVRT, is the most common PSVT — picking AVRT as 'most common' is the classic distractor. Also, the buried/pseudo-r' P wave (not absent P waves) distinguishes AVNRT from AF.

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5. AVRT runner + delta

WHAT YOU SEE

A runner labelled 'AVRT' beside a 'DELTA' arrow and 'SHORT PR' — the runner taking an extra side-track (accessory pathway) = the bypass route of AVRT, betrayed by a short PR and delta wave.

WHAT IT ENCODES

AV re-entrant tachycardia uses an accessory pathway (e.g., bundle of Kent) connecting atrium to ventricle outside the AV node. In sinus rhythm with WPW, antegrade conduction down the accessory pathway pre-excites the ventricle, giving a SHORT PR (<120 ms) and a DELTA wave (slurred QRS upstroke). During orthodromic AVRT (most common, ~95%) the circuit goes down the AV node and back up the accessory pathway, yielding a narrow QRS with a retrograde P after the QRS (RP < PR, but P separated from QRS).

BOARD TRAP

Antidromic AVRT conducts DOWN the accessory pathway, producing a WIDE, maximally pre-excited QRS that mimics VT — do not reflexively call it 'VT' or treat with standard AV-nodal blockers. Distinguish orthodromic (narrow) from antidromic (wide) AVRT.

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6. Delta = pre-excitation

WHAT YOU SEE

A 'DELTA' triangle painted on the track itself — the slurred ramp onto the rail = the delta wave, the slurred QRS upstroke of accessory-pathway pre-excitation (WPW).

WHAT IT ENCODES

The delta wave is the slurred initial QRS upstroke caused by accessory-pathway pre-excitation — the hallmark of Wolff-Parkinson-White on the resting ECG (short PR + delta wave + wide QRS). WPW syndrome = the ECG pattern PLUS symptomatic arrhythmia.

BOARD TRAP

The most feared scenario in WPW is PRE-EXCITED ATRIAL FIBRILLATION: rapid irregular wide-complex tachycardia. AV-nodal blockers (adenosine, verapamil/diltiazem, beta-blockers, digoxin) are CONTRAINDICATED because they shunt conduction down the accessory pathway and can accelerate to VF — use procainamide (or ibutilide) or synchronized cardioversion instead.

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7. Focal AT

WHAT YOU SEE

A single lone passenger labelled 'FOCAL AT' standing apart from the crowd — one rogue traveller = a single ectopic atrial focus firing on its own, faster than the sinus node.

WHAT IT ENCODES

Focal atrial tachycardia arises from a single ectopic atrial focus firing faster than the sinus node (automaticity/triggered activity). The P-wave morphology DIFFERS from sinus, there is an isoelectric baseline between P waves, and it is a long-RP tachycardia. It often shows a 'warm-up' (gradually accelerating) onset.

BOARD TRAP

Focal AT is AV-node-independent, so adenosine usually only causes transient AV block (revealing the abnormal P waves) rather than terminating it — unlike AVNRT/AVRT which it terminates.

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8. AT rate >200

WHAT YOU SEE

A '>200' sign on the platform — the posted high speed limit = the fast atrial rate of atrial tachycardia, with discrete P waves on a flat baseline.

WHAT IT ENCODES

Atrial tachycardias run at atrial rates of roughly 100-250 bpm (focal AT often 150-200+). Key feature: a clear isoelectric baseline between discrete P waves distinguishes AT from atrial flutter, where the undulating sawtooth has no isoelectric baseline (flutter atrial rate is classically ~300 bpm).

BOARD TRAP

Don't confuse atrial tachycardia with atrial flutter: flutter has continuous sawtooth waves (no isoelectric baseline) at ~300/min, whereas AT has discrete P waves separated by a flat baseline.

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9. Lungs (MAT)

WHAT YOU SEE

A pair of lungs labelled by the atrial-tachy platform — diseased lungs (severe COPD/hypoxia) = the classic setting for multifocal atrial tachycardia, with its ≥3 P-wave shapes.

WHAT IT ENCODES

Multifocal atrial tachycardia is defined by ≥3 distinct P-wave morphologies with varying PP, PR, and RR intervals — an irregular narrow-complex rhythm classically seen in severe COPD/hypoxia and also in theophylline toxicity and hypomagnesemia. Treatment targets the underlying lung disease and corrects hypoxia/electrolytes; verapamil or a beta-blocker (cautiously, given COPD) can be used for rate.

BOARD TRAP

MAT is irregularly irregular and is easily MISTAKEN for atrial fibrillation — the discriminator is the presence of discrete, multiform P waves in MAT (AF has no organized P waves). Also, avoid non-selective beta-blockers in bronchospastic COPD patients with MAT.

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10. Older adults

WHAT YOU SEE

An 'OLDER ADULTS' label on the atrial-tachy side — the elderly tag = the typical demographic for focal AT/MAT (structural heart/lung disease), versus young patients for AVNRT/AVRT.

WHAT IT ENCODES

Atrial tachycardias (focal AT, MAT) are more common in OLDER patients with structural heart disease, COPD, or critical illness — in contrast to AVNRT/AVRT, which classically present in younger patients with structurally normal hearts.

BOARD TRAP

Age/comorbidity helps pre-test probability: a young healthy patient with paroxysmal palpitations is more likely AVNRT/AVRT; an elderly COPD patient with irregular narrow tachy is more likely MAT — anchoring on the wrong demographic leads to the wrong diagnosis.

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11. Treat the cause

WHAT YOU SEE

A 'TREAT THE CAUSE' pennant by the sign-up desk — the banner reminds you automatic atrial rhythms (AT/MAT) are driven by reversible triggers (hypoxia, low K/Mg, dig toxicity) to fix.

WHAT IT ENCODES

Automatic atrial rhythms (focal AT, MAT) are often driven by reversible triggers — ischemia, hypoxia, sympathetic surge, electrolyte derangements (low K^+/Mg^{2+}), digoxin toxicity (classic for AT with block), and theophylline. Management emphasizes reversing the trigger, not just AV-nodal rate control.

BOARD TRAP

AT WITH BLOCK plus this picture should trigger thought of DIGOXIN TOXICITY — giving more AV-nodal blockade or ignoring the dig level is the trap; check the digoxin level and correct potassium.

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12. Wide = VT until proven

WHAT YOU SEE

A heavy locomotive charging in with red 'WIDE → VT' flags — the big wide engine = any wide-complex tachycardia, flagged as VT until proven otherwise.

WHAT IT ENCODES

Any WIDE-complex tachycardia (QRS ≥120 ms) is ventricular tachycardia until proven otherwise — assume the most dangerous diagnosis. Features favoring VT: AV dissociation, capture/fusion beats, very wide QRS (>140 ms in RBBB / >160 ms in LBBB morphology), extreme axis, concordance across precordial leads, and a history of structural heart disease/prior MI (the single best predictor).

BOARD TRAP

Defaulting to 'SVT with aberrancy' for a wide-complex tachycardia is a dangerous error — and giving verapamil to a misdiagnosed VT can cause hemodynamic collapse. When in doubt, treat as VT (procainamide/amiodarone if stable, cardiovert if unstable).

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13. Causes of wide QRS in SVT

WHAT YOU SEE

A note listing 'BBB / accessory pathway / drugs-K+' — the fine print explaining the few ways a supraventricular rhythm can masquerade as wide (aberrancy, pre-excitation, hyperK/Na-blockers).

WHAT IT ENCODES

An SVT can appear WIDE when there is: (1) a pre-existing or rate-related bundle branch block (aberrancy), (2) antegrade conduction over an accessory pathway (pre-excited AF or antidromic AVRT in WPW), or (3) drug/electrolyte effects that widen the QRS (hyperkalemia, sodium-channel blockers like TCAs/class I antiarrhythmics).

BOARD TRAP

Even knowing these causes, the safe default for an undifferentiated wide-complex tachycardia remains 'treat as VT' — only call it SVT with aberrancy when you have solid morphologic/clinical evidence.
